This tutorial, and MOG rasters for each state, are available on github:
https://github.com/Dan-Herrera/Mature-and-Old-Growth-Forest-Assessment

Assessing Mature and Old Growth Forest using FIA data

Mature and old growth forest support vast quantities of life, and are critical to the function of many ecosystems. While remotely sensed landcover datasets can indicate the spatial distribution of forests, they offer little insight to the age of forests. Fortunately, the United States Forest Service maintains an extensive forest inventory program across public and private lands called the Forest Inventory Analysis (FIA). This function uses R (specifically the RFIA package) to access FIA data from the Forest Service data repository, then assesses the data to determine if it meets the definitions for mature or old growth forest. Definitions vary by region and forest-type, offering substantial flexibility even across ecosystems. Finally, this function offers the option to interpolate the mature/old growth (MOG) status of unsampled forests based on neighboring sampled forests. The following tutorial explains how to use the function.

Identifying a study area

The `mapMOG()` function requires the user to identify a study area using a shapefile in sf format. The study area, called a locale, can be as large as a state, or as small as a forest. Please note, however, that the function cannot currently accommodate multi-state processing. If your study locale spans multiple states, the function must be run separately for each state contained in the locale. Additionally, small forests are not likely to contain enough FIA plots for adequate interpolation. Assessing data at the landscape-level (e.g., county, etc.) is likely to produce greater insights, even if you are only interested in a small portion of the landscape. In this tutorial we will use part of Olympic National Forest as a case study. A shapefile of Olympic National Forest is included in the function folder for your convenience.

Downloading and Accessing the function and data

Download the entire mapMOG folder from Dryad (file sizes exceed Github's limits) and unzip the folder if necessary. We recommend moving the folder from the Downloads folder of your computer to a more permanent location. ***Do not remove files from the mapMOG folder, as these files are necessary for proper functionality.***

File size limitations barred us from including all necessary files in the 'mapMOG' folder available for download. Prior to running the function, download the following files and add them to the mapMOG folder:

If you plan on interpolating MOG across unsampled landscapes

1. Download the most recent canopy cover layer from the National Land cover Database: https://www.mrlc.gov/data
2. Unzip the downloaded folder into the 'utility_canopy' subfolder of 'mapMOG' such that the file structure is '[YOUR FILE PATH]/mapMOG/utility_canopy/[CANOPY RASTER]'
3. No need to rename the raster

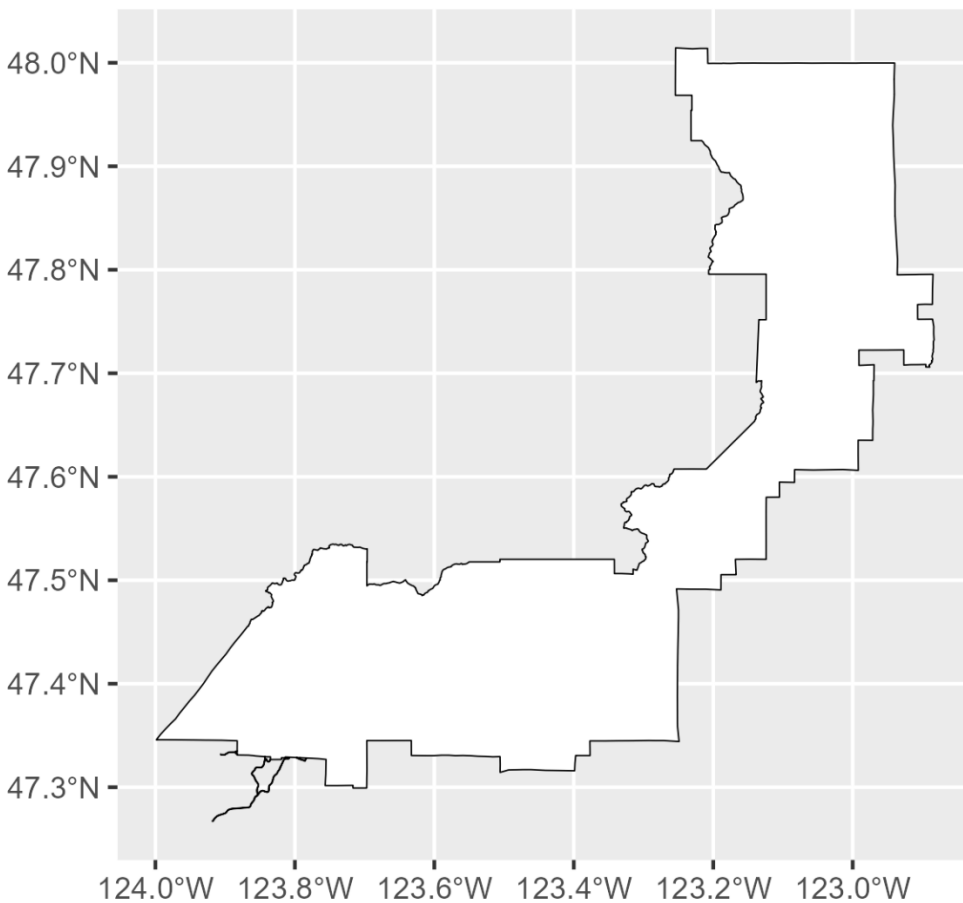
Once you have added the canopy raster to your `mapMOG` folder (if necessary), open R Studio and load the function by sourcing the script from the `mapMOG` folder. Note that the entire filepath must be specified for your computer to find the file.

```
> source("[YOUR FILE PATH]/mapMOG/FUNCTION_mapMOG.R") #loads function
into R environment
```

To access the Olympic National Forest shapefile, use the `getOlympic()` function which is now saved in your environment. If you are using your own shapefile, ensure that the file is in sf format. The function will reproject the shapefile as needed. I use ggplot2 to visualize shapefiles in this tutorial, but base R or other plotting packages can also be used.

```
> my.folderpath <- "[YOUR FILE PATH]/mapMOG" #save file path to mapMog
FOLDER (not function)

> olympicNF <- getOlympic(source.path = my.folderpath) #reads in Olympic
National Forest Shapefile

> library(ggplot2) #load ggplot2
```

```
> ggplot()+
   geom_sf(data = olympicNF, #plot the shapefile of Olympic National
Park
           fill = "white", #make the inside of the shape white
           color = "black")+ #make the outline of the shape black
   coord_sf(crs = 4326) #reproject so it is oriented correctly (WGS84)
```



Using the mapMOG function

Two arguments are needed to run the function: the shapefile of the locale, and the file path of the mapMOG folder. The recent and interpolate arguments both default to TRUE but can

be changed if desired. Only the most recent FIA plot data are used when `recent = TRUE`, and a raster of interpolated mature and old growth forests is produced when `interpolate = TRUE`. Finally, api defaults to the rFIA package. At this point, the rFIA package is the only supported API to access FIA data. Future versions of this function may incorporate alternative API's.

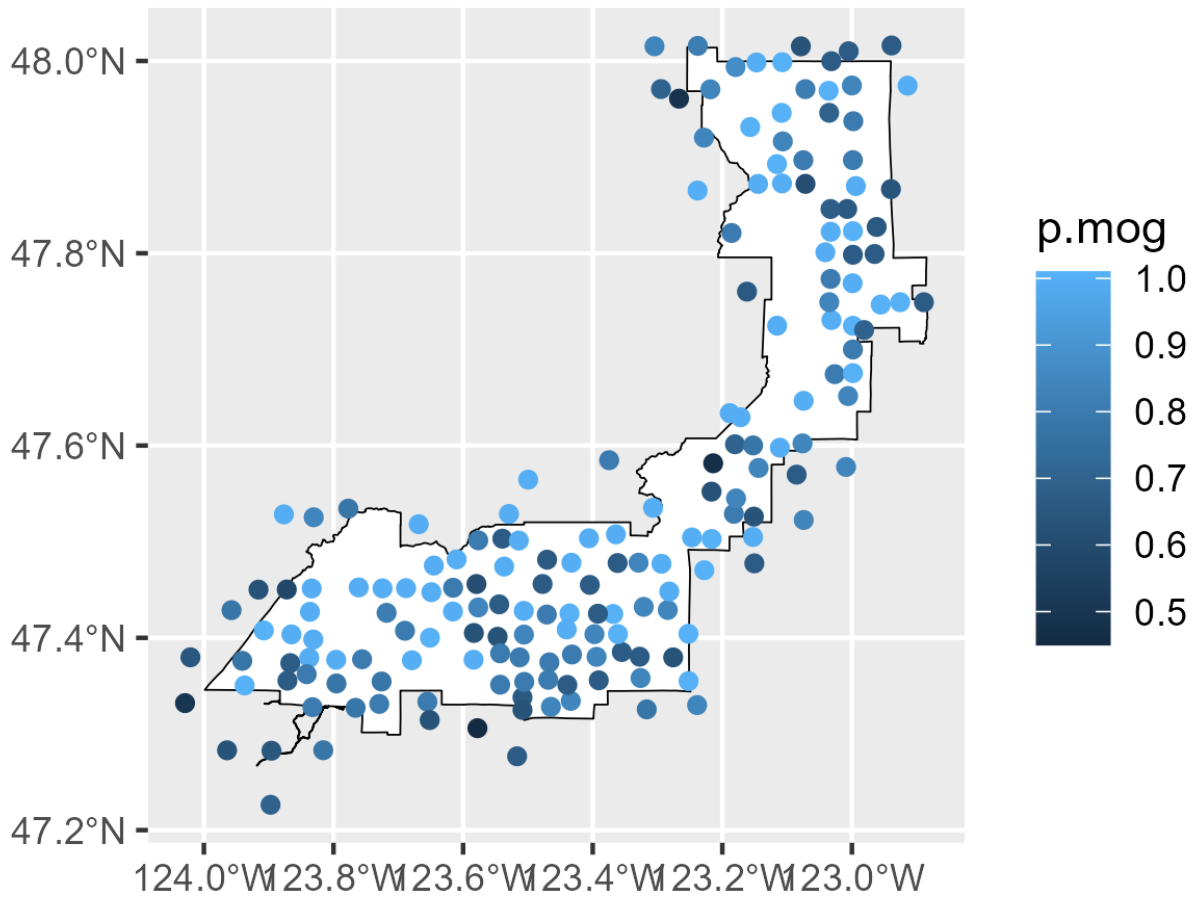
```
> mapMOG(locale = olympicNF, #specify your study area
        source.path = my.folderpath) #specify the location of the
folder that holds the function and auxiliary data
```

When you run the `mapMOG()` function, updates will print in your console ensuring that data has been successfully obtained. Error messages may also print if data are not available, etc. The `mapMOG()` function requires internet access to obtain data, but no error message will warn the user that the computer does not have access to internet. If an especially cryptic error message appears, check your internet connection.

Visualizing and interpreting the output

After the function is finished running, it will produce two outputs: a shapefile of FIA points classified by their MOG score, and a raster of interpolated MOG scores for all forested land in the locale. Note that these objects are automatically
named MOG.points and MOG.raster Let's take a look at the FIA points first:

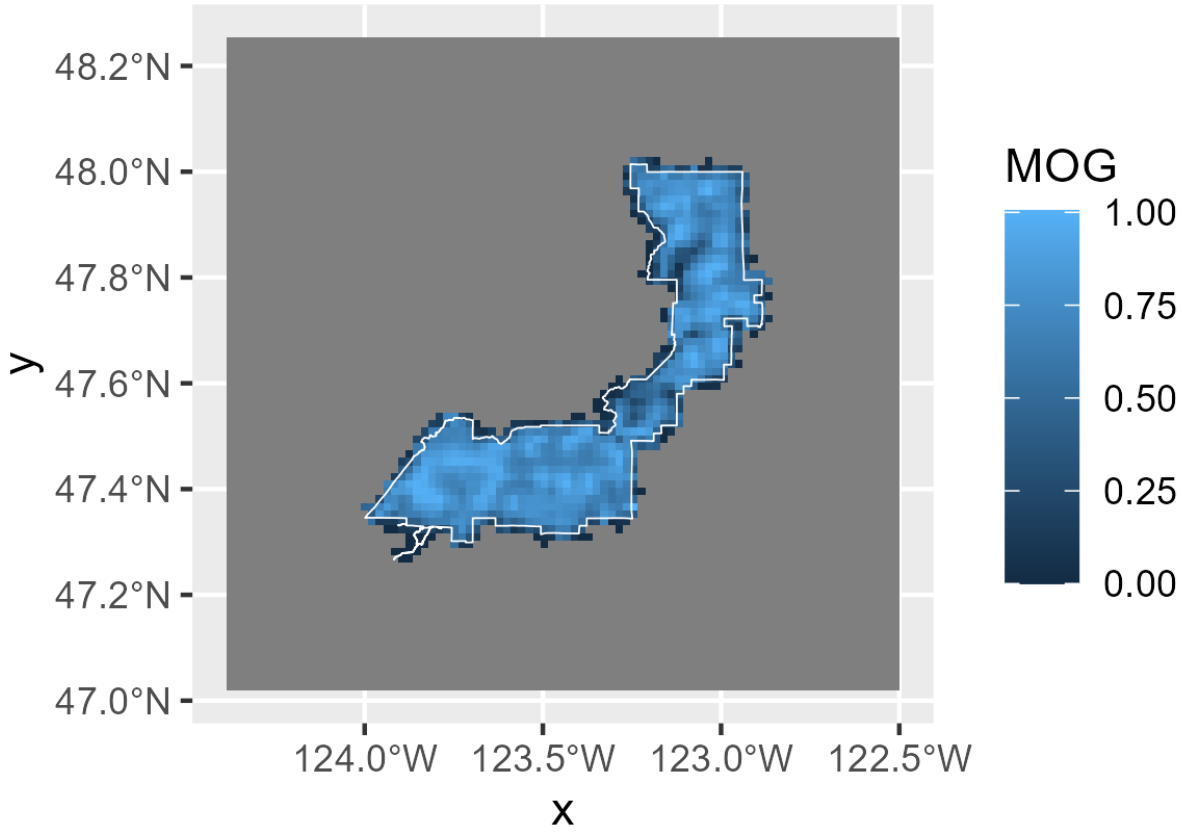
```
> ggplot()+
 geom_sf(data = olympicNF, #plot the shapefile of Olympic National Park
        fill = "white", #make the inside of the shape white
        color = "black")+ #make the outline of the shape black
 geom_sf(data = MOG.points, #plot FIA points
        aes(color = p.mog))+ #color FIA points by their MOG score
 coord_sf(crs = 4326) #reproject so it is oriented correctly (WGS84)
```



Each of the points on the map is a single FIA plot. The color of each point indicates the MOG score of that point. Scores range between zero and one, with zero indicating non-mature or old growth and one indicating old growth. A value of 0.5 indicates the forest is mature. In this example, all FIA plots in this section of Olympic National Forest are classified as being mature or old growth forest since all points have a score of 0.5 or greater. You may notice that some FIA points fall outside of the national forest. The function retrieves and processes data within 5 kilometers of the locale to minimize inaccurate interpolation at the locale edge due to sparse data.

You may notice that there are sections of the forest which do not have FIA data. We can use the interpolated MOG raster to offer insight about the MOG status of those areas based on the status of the three nearest FIA points. The fastest way to visualize this raster is to use base R: `plot(MOG.raster)`. However, for the sake of consistency, we will use ggplot2() in this example.

```
> #library(terra) #if you just ran the mapMOG function, the terra
package is already loaded. No need to load again


> MOG.raster.transformed <- terra::project(x = MOG.raster, #reproject
raster to ensure sensible plotting

                                  y = "epsg:4326") #WGS84

> plot.df <- as.data.frame(x = MOG.raster.transformed, #convert raster
into a dataframe

                         xy = TRUE) #save centroid coordinates of each
cell

> ggplot()+

  geom_tile(data = plot.df, #plot raster

              aes(x = x,

                  y = y,

                  fill = MOG))+ #color cells by MOG score

  geom_sf(data = olympicNF, #plot Olympic NF over raster to see
administrative boundaries

          fill = "transparent",

          color = "white")+

  coord_sf(crs = 4326) #reproject so it is oriented correctly (WGS84)
```



While interpolation can be helpful, it is important to remember that this is simply an estimation. Unsampled forest areas are estimated using inverse-distance weighting at a 1600x1600 meter (~1 mile) resolution. FIA plot data have slightly obscured geographic coordinates to protect landowner safety, and the interpolation process only occurs in raster cells that meet a canopy cover threshold specific to the locale.

Dynamic data exploration can be achieved using the mapview package: `mapview(MOG.raster.transformed)`

If you encounter issues with the mapMOG() function, please email Dan Herrera at *herrerawildlife(at)gmail.com*.